

IBM BoB AI INNOVATION HACKATHON 2026

Mission Readiness Officer

Predictive maintenance copilot for military fleets — powered by IBM BoB.

Team: CodeMasters Track: AI

Chetan Senta · github.com/ChetanSenta/bob-ai-hackathon-CodeMasters

FLEET STATUS

20 GREEN · 1 AMBER · 4 RED

TAIL-AH04 · AH-64 Apache

Rotor vibration 8.13 mm/s

READINESS 36/100 — RED

ASK BOB

“Why is TAIL-AH04 grounded?”

4 CRITICAL TASKS · 42.0h EST.

THE PROBLEM

Readiness data exists. Nobody is reading it in time.

\$90B

per year spent on
U.S. military maintenance

Fixed

servicing cycles — scheduled
by calendar, not condition

0

HUMS sensor streams
analysed in real time

!

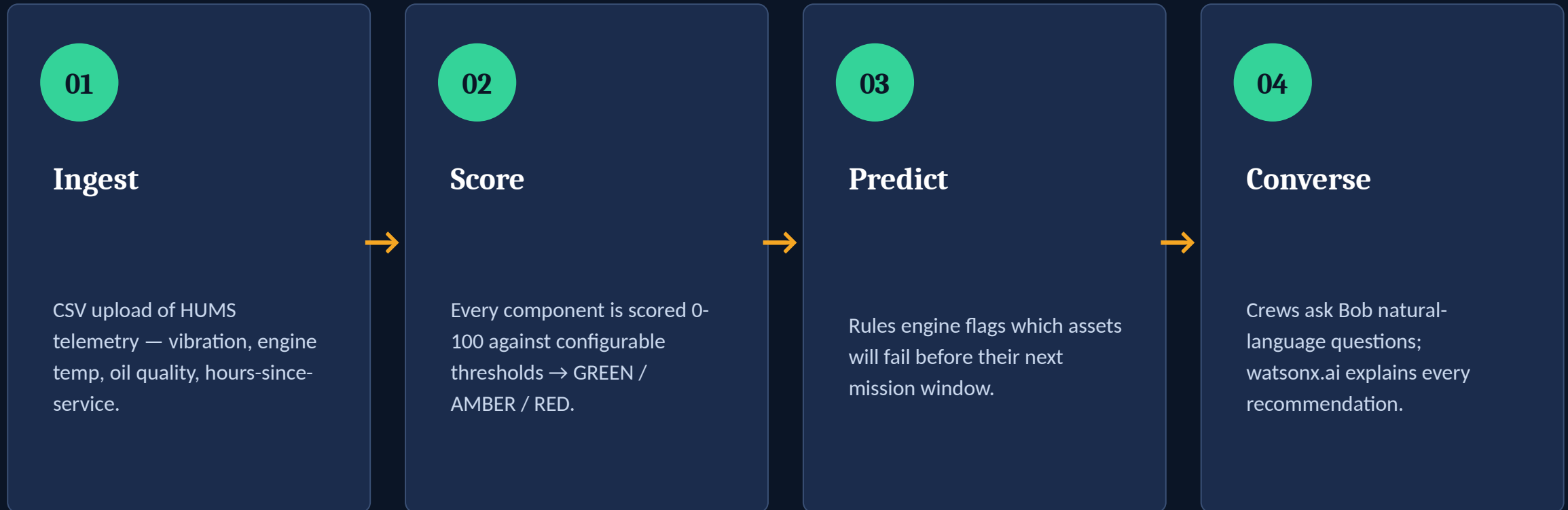
Result: unplanned groundings

Engines, rotors, and hydraulics generate HUMS telemetry every mission — but maintenance crews still work off fixed-schedule checklists. Failures surface at the worst possible moment: right before a mission window, when there's no time left to react. Every unexpected grounding is a readiness gap that a few hours of earlier warning could have prevented.

THE SOLUTION

The Mission Readiness Officer

A FastAPI rules engine + IBM BoB copilot that turns raw sensor data into a ranked, explainable maintenance plan.



KEY FEATURES

Built for the maintenance crew, not the dashboard

1

Fleet Readiness Dashboard

Real-time GREEN/AMBER/RED grid across the whole fleet, with per-component breakdown.

2

HUMS Data Ingestion

CSV upload for sensor readings and service records — no manual data entry.

3

Failure Prediction Engine

Rules-based component risk scoring, ranked by urgency and mission proximity.

4

IBM BoB Copilot

Conversational interface with tool-calling into four live backend endpoints.

5

watsonx.ai Explanations

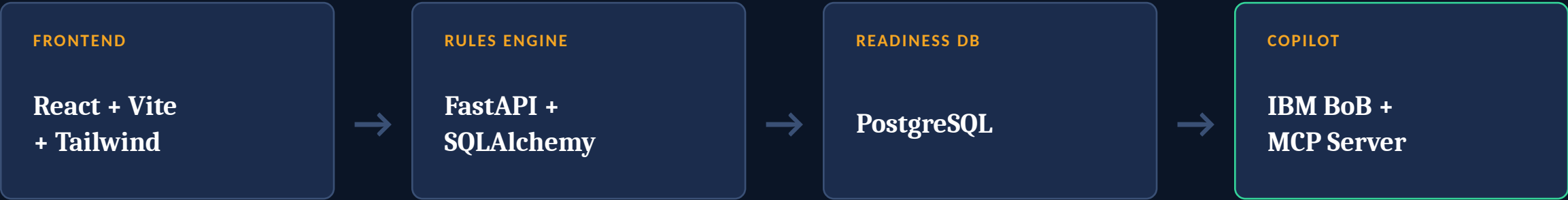
Plain-language maintenance recommendations for every AMBER/RED issue.

6

Prioritised Work Orders

Ranked task list with estimated hours and mission-window deadlines.

Tech stack & data flow



watsonx.ai called by the backend to generate natural-language readiness explanations for AMBER/RED assets.

Languages	Python, TypeScript
Frameworks	FastAPI, React, SQLAlchemy, Vite, Tailwind CSS
IBM Technologies	IBM BoB, watsonx.ai
Database	PostgreSQL
Infrastructure	Docker, Docker Compose, GitHub Actions

Bob is the mission interface, not a bolt-on

A real Node.js MCP server (src/mcp/) that Bob spawns as a child process, calling four typed tools that proxy live HTTP requests.

BOB MCP TOOL	BACKEND ENDPOINT	PURPOSE
get_fleet_readiness	GET /readiness/fleet	All assets with GREEN/AMBER/RED status
get_asset_detail	GET /readiness/asset/{id}	Per-component breakdown for one asset
get_failure_predictions	GET /predict/failures	Components at risk before next mission window
get_maintenance_plan	GET /maintenance/plan	Prioritised task list sorted by urgency

CONFIGURED IN .bob/

custom_modes.yaml

Defines the “Mission Readiness Officer” persona & tool rules

mcp.json

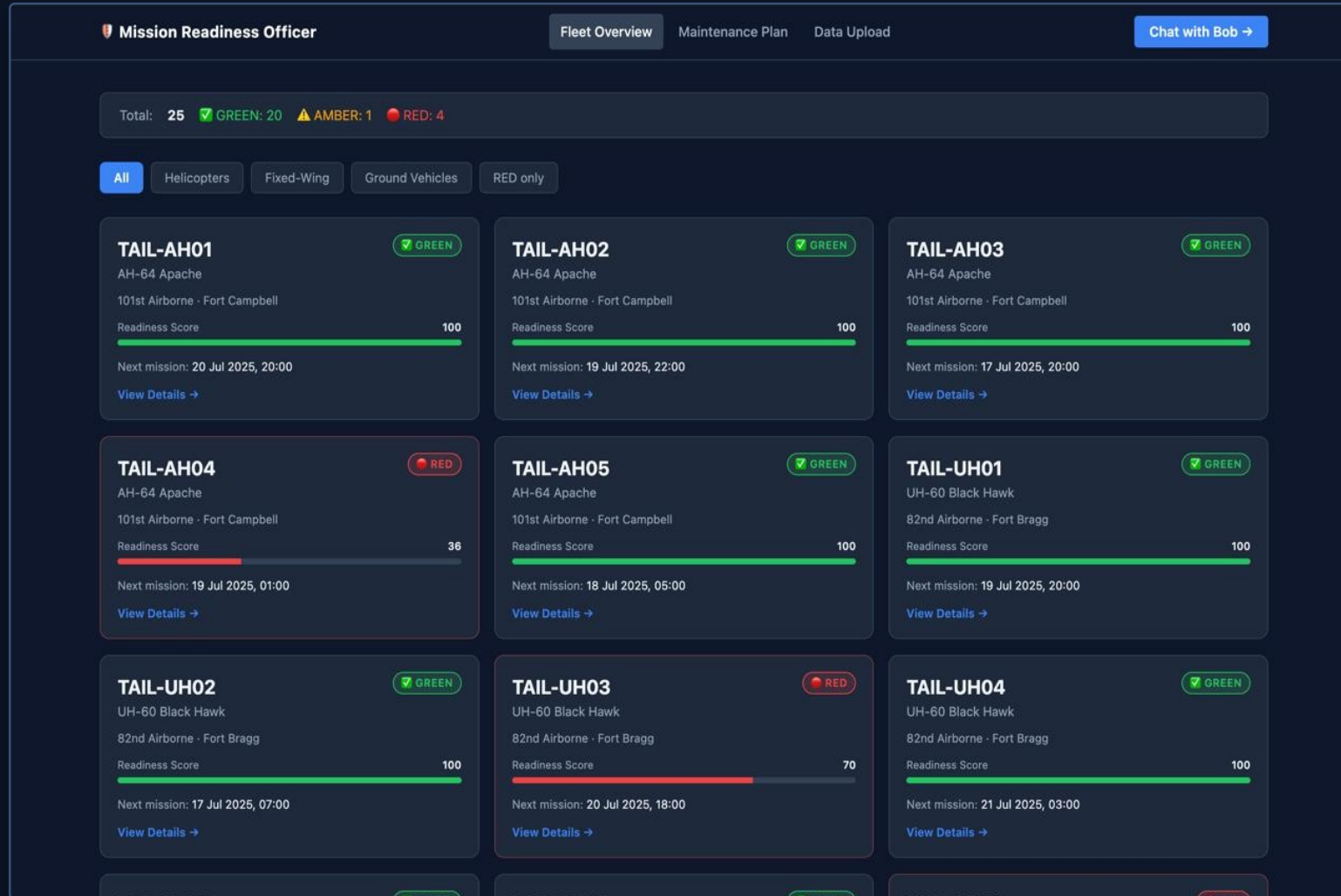
Registers the MCP server — Bob auto-connects on open

rules/mission-readiness.md

Domain context injected into every conversation

“Why is TAIL-AH04 grounded?”
“What will fail first?”
“Give me today's work orders”

Fleet readiness at a glance



25 assets, one screen

- Live GREEN / AMBER / RED status per tail number, filterable by asset class.
- Readiness score (0-100) and next mission window shown up front.
- One click into “View Details” for the full component breakdown.

At a glance: 4 assets RED, 1 AMBER — the plan on the next slide tells crews exactly what to do about it.

From one bad sensor reading to a work order

Mission Readiness Officer

Fleet OverviewMaintenance PlanData Upload

Chat with Bob →

TAIL-AH04

AH-64 Apache · helicopter · 101st Airborne · Fort Campbell

Next mission window: 19 Jul 2025, 01:00

Score: 36 / 100

Component Breakdown

Component	Metric	Value	Status	Score
rotor	vibration_mm_s	8.13	RED	36
airframe	hours_since_last_service	35.98	GREEN	100
hydraulics	oil_quality_index	85.1	GREEN	100
engine	engine_temp_c	155.14	GREEN	100

Readiness Assessment

READINESS ASSESSMENT — AH04 (AH-64 Apache) Overall Status: RED CRITICAL DEFICIENCIES: • ROTOR — vibration_mm_s = 8.13 (score 36/100). Immediate maintenance action required. Assessment generated by Mission Readiness Officer rules engine (template mode — watsonx.ai key not configured).

Recent Sensor Readings

Timestamp	Component	Metric	Value	Unit
13 Jul 2025, 19:04	rotor	vibration_mm_s	8.13	mm/s
13 Jul 2025, 19:04	airframe	hours_since_last_service	35.98	hours
13 Jul 2025, 19:04	hydraulics	oil_quality_index	85.1	index
13 Jul 2025, 19:04	engine	engine_temp_c	155.14	celsius

TAIL-AH04 — rotor vibration at 8.13 mm/s (threshold 7.0) drives the score to 36/100 and grounds the aircraft.

Mission Readiness Officer

Fleet OverviewMaintenance PlanData Upload

Chat with Bob →

Prioritised Maintenance Plan

Tuesday, 15 September 2026

4 CRITICAL

42.0h EST. TOTAL

6 TASKS

#	Urgency	Asset	Component	Action	Est. Hrs	Hrs Avail.	Mission Window
1	CRITICAL	TAIL-AW02 A-10 Thunderbolt II	hydraulics	Hydraulic oil flush and filter replacement	4h	-10199.6h	17 Jul 2025
2	CRITICAL	TAIL-AH04 AH-64 Apache	rotor	Rotor balance inspection and dynamic balancing	8h	-10166.6h	19 Jul 2025
3	CRITICAL	TAIL-MB03 M1 Abrams	drivetrain	Overdue scheduled maintenance — full component inspection required	10h	-10127.6h	20 Jul 2025
4	CRITICAL	TAIL-UH03 UH-60 Black Hawk	engine	Engine cooling system inspection and thermal regulation service	6h	-10125.6h	20 Jul 2025
5	HIGH	TAIL-CH04 C-130 Hercules	engine	Rotor balance inspection and dynamic balancing	8h	-10189.6h	18 Jul 2025
6	HIGH	TAIL-CH04 C-130 Hercules	engine	Engine cooling system inspection and thermal regulation service	6h	-10189.6h	18 Jul 2025

That single fault surfaces as a CRITICAL task — ranked, time-estimated, and tied to the mission window.

What's real, and what's next

WHAT WE'RE PROUD OF

Bob is the primary mission interface — not a cosmetic add-on. It spawns a real Node.js MCP server that proxies live HTTP calls to the FastAPI backend, then synthesises the JSON into concise, military-style briefings.

Remove Bob and analysts are left with only the dashboard; the conversational readiness interrogation disappears entirely.

KNOWN LIMITATIONS

- Synthetic dataset — not sourced from real HUMS hardware
- Rule-based scoring — hand-tuned thresholds, not a trained ML model
- watsonx.ai is optional — falls back to template text without an API key
- Single-node deployment — no auth layer or horizontal scaling; built for demo

MISSION READY.

Turning sensor data into decisions — before the mission window closes.

GREEN 20

AMBER 1

RED 4

Team CodeMasters

Chetan Senta

github.com/ChetanSenta/bob-ai-hackathon-CodeMasters